



Headaches and Migraines

How Do You Diagnose an Adult with a Migraine?

A patient must experience 5 or more headaches with at least 2 of the following features:

- **Nausea:** Is the patient nauseated or sick to their stomach when they have a headache?
- **Disability:** Has a headache limited a patient's activities for a day or more in the last 3 months?
- **Photophobia:** Does light bother the patient when they have a headache?

2 of 3 symptoms, PPV = 93%; 3 of 3 symptoms, PPV = 98%.



What are the Red Flag Symptoms?

S: Systemic symptoms (fever, weight loss)

N: Neuro symptoms (confusion, altered level of consciousness)

O: Onset (sudden, abrupt, or split-second; time to peak 1 minute or less)

O: Older patient (new onset or progressive, especially if > 50 years)

P: Previous history of headache different? (Change in clinical features or abrupt changes in frequency or severity with new symptoms that are not typical for the person.)

P: Positional or postural (headache only when supine or only when standing)

P: Precipitated by cough, sneeze, or Valsalva

P: Papilledema

S: Secondary risk factors (HIV, systemic cancer)

How to Choose Acute Migraine Medication



Note: From Sumatriptan [Photograph], by GoodRx, n.d., GoodRx (<https://www.goodrx.com/sumatriptan/what-is>)

- **Sumatriptan 100 mg** is the preferred drug.
- **Headache recurs the same day:** Switch to rizatriptan.
- **Sumatriptan side effects:** Switch to naratriptan or sumatriptan 50 mg.
- **Nausea/vomiting:** ADD antiemetic at onset of headache OR try sumatriptan injectable.
- **Wakes up with severe migraine:** Use sumatriptan injectable.
- For patients with inadequate relief from a triptan, try combination therapy (triptans and NSAID and/or anti-emetics).

Tips for Success

- Treat early in attack.
- Treat at least three different attacks with same medication before determining efficacy.
- Use acute meds for maximum of 2-3 days per week, no more than 9 days per month.
- Encourage management of triggers (caffeine, sleep deprivation, dehydration, ETOH, fasting or skipping meals, etc.)

Avoid Medication Overuse Headaches

- No more than 14 days of any pain reliever (including OTC) per month.
- No more than 9 days of triptan use per month.

Contraindications for Use of Triptans

- Arrhythmias associated with cardiac accessory conduction pathway disorders, including Wolff-Parkinson-White syndrome.
- Concomitant administration of MAO-A inhibitors or use within 2 weeks of discontinuation of MAO-A inhibitor therapy.
- Concomitant use of ergotamine-containing or ergot-type medication (e.g., dihydroergotamine, methysergide) within 24 hours.
- Concomitant use with another 5-hydroxytryptamine(1) agonist within 24 hours.
- Coronary artery vasospasm, including Prinzmetal angina, or other significant underlying cardiovascular disease.
- History of hemiplegic or basilar migraine.
- History of stroke or transient ischemic attack.
- Hypersensitivity to sumatriptan, or any of its components.
- Ischemic bowel disease.
- Ischemic coronary artery disease (angina pectoris, history of myocardial infarction, or documented silent ischemia).
- Peripheral vascular disease.
- Severe hepatic impairment.
- Uncontrolled hypertension.

Distinguishing Migraines from Headaches

Migraines v. Tension-Type Headaches?

Migraines will include: nausea and/or photophobia, phonophobia, and/or exacerbation by physical activity.

Tension-type headaches will not.

Sinus Headaches v. Migraines

- Migraine is commonly misdiagnosed as sinus headache.
- Self-diagnosed sinus headache is nearly always migraine (~90% of the time).
- Refer to the Sinusitis SmartRx of GL on the clinical library for more information.



Note: From Photophobia [Photograph], by American Migraine Foundation, 2017, American Migraine Foundation (<https://americanmigrainefoundation.org/resource-library/photophobia-migraine/>)

Migraine Prophylaxis

Criteria for Migraine Prophylaxis

1. >=8 headache days per month.
2. >=2 disabling migraine attacks per month that last >=3 days.
3. Other reasons
 - Infrequent headache attacks that produce profound disability.
 - Failure of, or contraindication to, acute medications.
 - Overuse of acute medications.
 - Patient preference.
 - Pattern of increasing attacks over time.

Prophylactic Treatment

Expectations: Success is a 59% reduction in frequency and intensity of the HA. Adverse effects can often be self-limited and dose-dependent.

Avoid narcotics and butalbital combination products.

If the following **comorbidities** are present, consider these migraine prophylactic options:

- **Depression/anxiety, other chronic pain syndrome:** TCA (if age <65), venlafaxine.
- **Insomnia:** Cyproheptadine (if age <65), TCA (if age <65), venlafaxine.
- **Pregnancy potential:** Avoid valproate, topiramate.
- **Currently pregnant:** Use magnesium (avoid valproate, topiramate, feverfew)
- **Obesity:** Topiramate (avoid valproate, cyproheptadine, possibly TCAs)
- **Poor appetite/FTT:** Valproate, cyproheptadine (avoid topiramate).
- **Seizures:** Topiramate or valproate.
- **Prolonged aura, hemiplegia:** Verapamil.
- **On SSRI or SNRI:** Avoid cyproheptadine.
- **Patient strongly prefers "natural" remedies:** Riboflavin, feverfew, magnesium (avoid butterbur).

Dose Optimization

- Increase over 4 - 6 weeks.
- Evaluate efficacy after 6 weeks.
- Follow-up at 6-months on optimal dose is ideal to determine success.

Partial response but not 50% reduction in HA days/month after 8 - 12 weeks

- **Combine** current treatment with riboflavin, magnesium, and/or feverfew.

No improvement after 8 -12 weeks? Switch to another migraine preventative.

Drug	Titration	Optimal Dose
TCA (amit, nortrip <65 yrs)	10 - 25 mg/dose every 7-14 days	30 - 150 mg/day
Venlafaxine	37.5 - 75 mg/dose weekly	150 mg/day
Propranolol	20 - 40 mg/dose weekly	120 - 140 mg/day
Metoprolol	25 - 50 mg/dose weekly	200 mg/day
Atenolol	25 - 50 mg/dose weekly	100 mg/day
Topiramate	25 mg/dose every 4 - 7 days	100 mg/day (once daily to BID)
Divalproex EC/DR	250 - 500 mg/dose every 4 - 7 days	500 - 1,500 mg/day
Verapamil	120 mg SR/dose weekly by HR greater	240 - 480 mg SR daily (BID)
Divalproex EC/DR	250 - 500 mg/dose every 4 - 7 days	500 - 1,500 mg/day

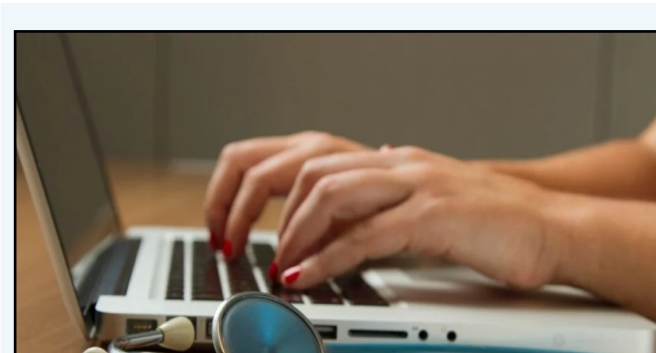
Alternative Medications

Drug	Optimal Dose
Riboflavin	400 mg once daily (Level B Evidence)
Magnesium	400 mg once daily (Level B Evidence)
Feverfew leaves/capsules	50 - 100 mg once daily
Feverfew (MIG-99) CO2 extract	6.25 mg TID (Level B Evidence)

Prophylactic Pearls

1. Full benefit may **not** be seen until 6 months have elapsed on optimal dose.
2. Reinforce education and lifestyle management and behavioral therapies.
3. Encourage management of triggers (caffeine, sleep deprivation, ETOH, fasting or skipping meals, etc.).

Introducing the GPC Mailbag!



At GPC Clinical Corner, we strive to provide tips on the best clinical practices, navigating patient encounters, and updated information on treatment options.

This year, we'd like to expand our scope to include your suggestions. We're proud to announce the GPC Mailbag, where you can request topics for discussion, or even specific questions, that you'd like to see in a future newsletter. You can reach us at GPCMailbag@kp.org. We look forward to hearing from you!

References

American Migraine Foundation. (2017). *Photophobia* [Photograph]. American Migraine Foundation. <https://americanmigrainefoundation.org/resource-library/photophobia-migraine/>

GoodRx. (n.d.) *Sumatriptan* [Photograph]. GoodRx. <https://www.goodrx.com/sumatriptan/what-is>